

Combined Immunodeficiency Due to MALT1 Deficiency (IMD12): A Comprehensive Disease Characterization

Disease: Combined Immunodeficiency Due to MALT1 Deficiency **MONDO ID:** MONDO:0014197 | **OMIM:** #615468 (Immunodeficiency-12, IMD12) | **Gene:** *MALT1* (HGNC:6819; OMIM 604860)
*Category:** Mendelian (autosomal recessive inborn error of immunity)

Summary

Combined immunodeficiency due to MALT1 deficiency, also designated **Immunodeficiency-12 (IMD12; OMIM #615468; MONDO:0014197)**, is an ultra-rare autosomal recessive inborn error of immunity caused by biallelic loss-of-function variants in *MALT1*. MALT1 (mucosa-associated lymphoid tissue lymphoma translocation protein 1) is the only human paracaspase and serves as a core subunit of the **CARD11–BCL10–MALT1 (CBM) signalosome**, the molecular hub that couples antigen-receptor engagement on T and B lymphocytes to canonical NF-κB activation. When MALT1 function is lost, the CBM complex can no longer transmit antigen-receptor signals to the threshold required for effective lymphocyte activation, producing a disease that combines features of classical immunodeficiency (recurrent infection, poor antibody responses) with immune dysregulation (eczema, elevated IgE, eosinophilia, autoimmunity) — a duality that stems directly from MALT1's dual role in both effector-lymphocyte activation and regulatory T-cell (Treg) homeostasis.

Clinically, affected infants present in the first weeks to months of life (mean onset ~1.6 months) with recurrent bacterial, viral, and fungal infections (100%), skin involvement/eczema (100%), failure to thrive (100%), oral lesions (67%), chronic diarrhea (56%), and autoimmunity (44%). The characteristic laboratory signature includes eosinophilia (~67%), elevated IgE (~22%), typically normal T and NK cell counts but reduced B cells (89% of patients), arrest of B-cell maturation, and impaired specific antibody responses. The disorder is heavily enriched in consanguineous Middle Eastern, Turkish, and North African populations, consistent with a consanguinity/founder effect rather than a single global founder allele, and fewer than ~20 patients had been comprehensively characterized as of 2022.

The prognosis is poor with conservative management: immunoglobulin replacement and antimicrobial prophylaxis are largely ineffective, and infection-related mortality is high (5 of 9 patients died in the largest cohort). **Allogeneic hematopoietic stem cell transplantation (HSCT)** — which reconstitutes CBM-competent hematopoietic cells — is the only established curative therapy. MALT1 deficiency sits within the broader family of "**CBM-opathies**" (inborn errors of CARD11, BCL10, and MALT1), and its eczema/high-IgE/sinopulmonary-infection phenotype overlaps with DOCK8 deficiency and hyper-IgE syndromes, so molecular confirmation is required for diagnosis.

1. Disease Information

MALT1 deficiency is an autosomal recessive **combined immunodeficiency (CID)** with early-onset multisystem disease. It is formally classified as **Immunodeficiency-12 with susceptibility to viral, bacterial, and fungal infections (IMD12)**.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0014197
OMIM (phenotype)	#615468 (IMD12)
OMIM (gene)	604860 (MALT1*)
HGNC	HGNC:6819
Orphanet	Classified under combined immunodeficiencies (prevalence <1/1,000,000)
Gene locus	18q21.32

Synonyms / alternative names: Immunodeficiency-12; IMD12; MALT1 deficiency; MALT1 paracaspase deficiency; combined immunodeficiency due to MALT1 deficiency; CBM-opathy (MALT1 type).

Information source: The disease-level characterization derives from aggregated case series and individual case reports in the primary literature (e.g., the 19-patient synthesis in [PMID: 35079916](#)), supplemented by mechanistic studies in cell lines and model organisms. Given extreme rarity, there are no EHR-scale or registry-scale datasets; all clinical data are individual-patient-derived and then aggregated.

2. Etiology

Primary cause — genetic: The disease is caused by **biallelic (homozygous or compound heterozygous) loss-of-function variants in *MALT1***. In consanguineous families the variants are typically homozygous. There is no environmental or infectious cause of the underlying disorder; infections are the *consequence* of the immune defect, not its cause.

Genetic risk factors: The sole causal factor is biallelic *MALT1* LOF. **Consanguinity** is the dominant population-level risk factor, as it dramatically increases the probability of homozygosity for rare recessive alleles. No modifier genes have been formally validated, though the broader severity of CBM-opathies depends on which component and residual protein function is affected.

Environmental / lifestyle risk factors: None established as causal. Pathogen exposure determines the pattern and timing of clinical infection but does not cause the immunodeficiency.

Protective factors: Heterozygous carriers are asymptomatic (one functional allele suffices). No protective modifier alleles have been described. The only "protective" intervention is definitive immune reconstitution via HSCT.

Gene–environment interactions: The genetic lesion sets a lowered threshold for immune failure; environmental pathogen load then determines clinical expression (severity/frequency of infections, triggering of eczema flares). This is a "genotype sets susceptibility, environment determines manifestation" relationship rather than a true molecular GxE interaction.

3. Phenotypes

Phenotype frequencies are drawn primarily from the largest cohort (9 new + 10 previously reported patients, n=19; [PMID: 35079916](#)).

Phenotype	Type	Frequency	Suggested HPO term
Recurrent infections (bacterial/viral/fungal)	Clinical sign	100%	HP:0002719 (Recurrent infections)
Skin involvement / eczema / dermatitis	Physical manifestation	100%	HP:0000964 (Eczema) / HP:0000988 (Skin rash)
Failure to thrive	Clinical sign	100%	HP:0001508 (Failure to thrive)
Oral lesions (ulcers, candidiasis)	Clinical sign	67%	HP:0000155 (Oral ulcer)
Chronic diarrhea / enteropathy	Symptom	56%	HP:0002028 (Chronic diarrhea)
Autoimmunity	Clinical sign	44%	HP:0002960 (Autoimmunity)
Eosinophilia	Lab abnormality	67%	HP:0001880 (Eosinophilia)
Elevated serum IgE	Lab abnormality	22%	HP:0003212 (Increased IgE level)
Reduced B cells	Lab abnormality	89% (8/9)	HP:0010976 (B lymphocytopenia)
B-cell maturation arrest	Lab abnormality	Reported in cases	HP:0005365 (Abnormal B cell morphology)
Impaired specific antibody responses	Lab abnormality	Common	HP:0005387 (Reduced antibody responses)

The phenotype frequencies come directly from the largest reported cohort ([PMID: 35079916](#)): *"The main clinical findings of the disease were recurrent infections (100%), skin involvement (100%), failure to thrive (100%), oral lesions (67%), chronic diarrhea (56%), and autoimmunity (44%)."*

Phenotype characteristics: - **Age of onset:** Neonatal / early infancy — mean disease onset **1.6 ± 0.7 months** (PMID: 35079916): *"The mean age of patients and disease onset were 33 ± 17 and 1.6 ± 0.7 months, respectively."* - **Severity:** Severe; life-threatening. - **Progression:** Progressive without curative treatment; chronic/lifelong course punctuated by acute infectious episodes. - **Immunophenotype:** *"The majority of patients had normal T and NK cells, while eight (89%) exhibited reduced B cells."* (PMID: 35079916)

Quality of life impact: Severe. Recurrent infections, chronic diarrhea/enteropathy, failure to thrive, and dermatitis substantially impair growth, nutrition, and daily function in infancy; the burden of hospitalization and the need for HSCT are profound. Formal QoL instrument data (EQ-5D, SF-36) are not available for this ultra-rare disorder.

4. Genetic / Molecular Information

Causal gene: *MALT1* (HGNC:6819; chromosome **18q21.32**; OMIM 604860). *The gene encodes an 824-amino-acid paracaspase** — the only paracaspase in humans.

Protein architecture (relevant to loss-of-function mechanism): - N-terminal **death domain** - Two N-terminal **Ig-like (immunoglobulin-like) domains** (Ig1–Ig2; mediate BCL10 binding and oligomerization) - Central **caspase-like (paracaspase) protease domain** - C-terminal **Ig-like domain**

Crystallography (1.75 Å) shows the paracaspase domain adopts a fold nearly identical to classic caspases and homodimerizes to form an active protease that cleaves substrates after arginine residues (PMID: 22158899): *"The paracaspase domain adopts a fold that is nearly identical to that of classic caspases and homodimerizes similarly to form an active protease."* The tandem Ig-like domains additionally mediate oligomerization important for signaling (PMID: 21966355).

Reported pathogenic variants (all germline, biallelic, loss-of-function):

Variant (cDNA / protein)	Domain	Type	Reference
c.1411G>A; p.D471N	caspase-like	Missense (LOF)	PMID: 40748513
c.762dup; p.Ile255TyrfsTer10	N-terminal	Frameshift	PMID: 39017781
p.K543R + p.M732T (compound het)	caspase-like / C-terminal	Hypomorphic missense	PMID: 41882201

Supporting quotes: - *"The patient carried a novel pathogenic biallelic loss-of-function variant in MALT1 (c.1411G > A; p.D471N) located in the caspase-like domain, leading to severely reduced MALT1 protein expression."* ([PMID: 40748513](#)) - *"Next-generation sequencing revealed a novel homozygous variant in the MALT1 gene (c.762dup in exon 5 of 17; p.Ile255TyrfsTer10); this variant is likely pathogenic, thus supporting the genetic diagnosis of immunodeficiency-12 (IMD12)."* ([PMID: 39017781](#)) - *"the MALT1 K543R and M732T variants attenuated MALT1's enzymatic activity and compromised its protein stability"* ([PMID: 41882201](#))

Variant classification: Pathogenic / likely pathogenic per ACMG/AMP, supported by functional assays demonstrating reduced protein and/or reduced protease/scaffold activity.

Variant types: Missense, frameshift, and nonsense have all been reported. **Functional consequence = loss of function** (reduced protein stability and/or reduced protease and scaffold activity).

Allele frequency: Pathogenic *MALT1* LOF alleles are essentially absent in the homozygous state in gnomAD, consistent with recessive severe disease.

Somatic vs germline: Disease-causing variants are **germline**. (Note: *MALT1* is also oncogenically activated in lymphoma via the somatic t(11;18) API2-MALT1 fusion and chronic CBM activation — a distinct, unrelated pathology.)

Modifier genes / epigenetics / chromosomal abnormalities: No validated modifier genes, disease-specific epigenetic signatures, or large-scale chromosomal abnormalities are described for the Mendelian deficiency.

5. Environmental Information

Environmental factors: No toxic, radiation, or occupational exposures contribute to this monogenic disease.

Lifestyle factors: Not applicable (disease of infancy).

Infectious agents: Pathogens are **downstream consequences**, not causes. The combined immunodeficiency predisposes to recurrent bacterial, viral, and fungal infections. Staphylococcal skin/soft-tissue infections and mucocutaneous candidiasis are notable, overlapping with the DOCK8/hyper-IgE differential ([PMID: 39017781](#)). Chronic mucocutaneous candidiasis-type and other opportunistic infections reflect the combined T/B defect.

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic loss-of-function variants in *MALT1*** (missense/frameshift/nonsense) **lead to** severely reduced MALT1 protein and/or loss of paracaspase protease and scaffold activity.
2. Loss of functional MALT1 **results in** a non-functional or crippled **CARD11–BCL10–MALT1 (CBM) signalosome**, which normally assembles downstream of the T-cell receptor (TCR) and B-cell receptor (BCR).
3. A defective CBM complex **fails to transmit antigen-receptor signals** to the threshold required for **canonical NF- κ B activation** (demonstrated by reduced p65/RelA phosphorylation).
4. Impaired NF- κ B signaling **branches** into two consequences:
5. **(Branch A — effector/immunodeficiency):** reduced lymphocyte activation, proliferation, and cytokine output (deficient IL-2 and TNF- α), **leading to** impaired T-cell help, arrest of B-cell maturation at the transitional/naïve stage, reduced memory and total B cells, and poor specific antibody responses → **recurrent infections and (functional) agammaglobulinemia**.
6. **(Branch B — regulatory/dysregulation):** loss of MALT1 protease activity **results in** reduced regulatory T-cell (Treg) and follicular regulatory (Tfr) cell numbers and function, plus Th2 skewing, **leading to** autoimmunity, eczema/atopic dermatitis, eosinophilia, and elevated IgE.

7. The combined effector failure plus immune dysregulation **manifests clinically** as early-infancy combined immunodeficiency with infections, enteropathy, failure to thrive, dermatitis, and autoimmunity.

(The branching in step 4 is directly inferred from the dual scaffold/protease biochemistry of MALT1 and is demonstrated in both patient cells and model organisms; see below.)

Molecular and cellular detail

Molecular pathway (central): The CBM signalosome bridges ITAM-coupled antigen receptors (TCR/BCR) to **NF-κB**, and also to **JNK** and **mTORC1** (PMID: 30283440). MALT1 integrates receptor-derived signals and *"determines whether downstream nuclear factor kappa B (NF-κB) activation reaches thresholds required for effective immune responses"* (PMID: 42745540): *"As a central component of the CARD11-BCL10-MALT1 (CBM) signalosome, MALT1 integrates receptor-derived signals and determines whether downstream nuclear factor kappa B (NF-κB) activation reaches thresholds required for effective immune responses."*

Dual function of MALT1 (PMID: 37126937): *"MALT1 acts as a scaffolding protein to drive activation of NF-κB transcription factors and as a protease to modulate signaling and immune activation by cleavage of distinct substrates."* - **Scaffold function:** nucleates IKK activation → NF-κB. - **Protease function:** cleaves negative regulators of NF-κB (**A20**, **CYLD**, **RelB**) and RNA-binding repressors (**Regnase-1**, **Roquin**), thereby sustaining lymphocyte proliferation/survival and shaping cytokine mRNA stability.

Patient-level evidence: A p.D471N LOF variant *"Impaired CBM-mediated NF-κB activation was confirmed by reduced phosphorylation of the p65 subunit, resulting in deficient production of IL-2 and TNF-α"* and *"This functional defect caused lower Tfr and Treg cells, a normal proportion of Tfh cells, with higher expression of activation markers PD-1 and ICOS"* (PMID: 40748513). B-cell maturation arrest with elevated IgE and otherwise normal subset counts has been documented (PMID: 39017781): *"He had elevated serum IgE and normal B- and T-lymphocyte subset counts, but there was an arrest in the B-cell maturation."*

Treg dependence on MALT1 protease — mechanistic link (model organisms): MALT1 protease activity in Tregs drives TCR-induced upregulation of **MYC**, supporting mitochondrial function and homeostatic Treg expansion (PMID: 34668583): *"MALT1 protease activity controls the TCR-induced upregulation of the transcription factor MYC and the subsequent expression of MYC target genes in Tregs."* Thus the dysregulation branch is tied specifically to protease (not scaffold) function.

Cell types involved (suggested CL terms): CD4+ T cell (CL:0000624), CD8+ T cell (CL:0000625), regulatory T cell (CL:0000815), follicular regulatory T cell, B cell (CL:0000236), naïve/transitional B cell, memory B cell (CL:0000787), NK cell (CL:0000623), myeloid dendritic cell.

Suggested GO terms: antigen receptor-mediated signaling pathway (GO:0050851), I- κ B kinase/NF- κ B signaling (GO:0007249), positive regulation of NF- κ B transcription factor activity (GO:0051092), T cell receptor signaling pathway (GO:0050852), B cell receptor signaling pathway (GO:0050853), regulatory T cell differentiation (GO:0045066), proteolysis (GO:0006508), cysteine-type endopeptidase activity (GO:0004197).

Subcellular compartments (GO Cellular Component): cytoplasm/cytosol (GO:0005829) where the CBM signalosome assembles; the signal terminates in the nucleus (GO:0005634) via NF- κ B nuclear translocation.

Immune system involvement: Combined (T + B) immunodeficiency plus immune dysregulation (autoimmunity, atopy). This is the defining pathophysiology.

7. Anatomical Structures Affected

Organ / system level: - **Immune/lymphoreticular system** (UBERON:0002405 immune system; UBERON:0000029 lymph node; UBERON:0002106 spleen; UBERON:0002370 thymus) — primary. - **Skin** (UBERON:0002097) — eczema/dermatitis, staphylococcal infection. - **Gastrointestinal tract** (UBERON:0001555 digestive tract) — chronic diarrhea/enteropathy. - **Oral cavity / mucosa** (UBERON:0000167) — oral lesions, candidiasis. - **Respiratory tract** (UBERON:0000065) — recurrent sinopulmonary infection. - Systemic: failure to thrive reflects multi-organ nutritional/growth impact.

Tissue and cell level: Lymphoid tissue and peripheral blood leukocytes. Affected cell populations: T cells (including Tregs and Tfr), B cells (maturation arrest at transitional/naïve stage; reduced memory B cells), NK cells (numerically normal but functionally impaired), and myeloid dendritic cells.

Subcellular level: Cytoplasmic CBM signalosome assembly (GO:0005829); nuclear NF- κ B-driven transcription (GO:0005634).

Localization / lateralization: Systemic and bilateral/diffuse (not a focal or lateralized disease).

8. Temporal Development

- **Onset:** Congenital genetic defect with clinical onset in the **neonatal period / early infancy** (mean 1.6 ± 0.7 months; [PMID: 35079916](#)). Onset pattern is early and progressive.
 - **Progression:** Progressive and life-threatening without immune reconstitution; chronic/lifelong. Clinical course is punctuated by **acute infectious episodes** superimposed on chronic enteropathy, dermatitis, and failure to thrive.
 - **Disease course pattern:** Chronic-progressive with episodic infectious exacerbations.
 - **Remission:** No spontaneous remission. Durable remission/cure is achieved only through allogeneic HSCT.
 - **Critical period / intervention window:** Early infancy — prompt molecular diagnosis and HSCT before irreversible infectious/organ damage is the key window of opportunity.
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9. Inheritance and Population

Inheritance: Autosomal recessive. Affected individuals are homozygous (consanguineous families) or compound heterozygous for *MALT1* LOF variants; heterozygous carriers are asymptomatic.

Epidemiology: Ultra-rare. Only ~19 patients had been comprehensively characterized by 2022 (9 new + 10 previously reported; [PMID: 35079916](#)): *"We also analyzed ten previously reported patients to comprehensively evaluate genotype/phenotype correlation."* Additional single case reports have appeared since (Egyptian, Chinese patients). Orphanet lists prevalence as <1/1,000,000. No validated incidence figures exist.

Penetrance / expressivity: Biallelic LOF appears highly penetrant for combined immunodeficiency; expressivity is somewhat variable in the balance of immunodeficiency vs. dysregulation features and in residual B/T function, likely reflecting variant-specific residual activity (hypomorphic vs. null).

Founder effects / consanguinity: Reported families are frequently consanguineous with homozygous variants, heavily weighted toward **Middle Eastern/Turkish and North African** populations — consistent with a consanguinity/founder effect rather than a single global founder allele ([PMID: 39017781](#): *"Next-generation sequencing revealed a novel homozygous variant in the MALT1 gene"*).

Carrier frequency: Not formally established; pathogenic LOF alleles are essentially absent in the homozygous state in gnomAD.

Sex ratio / age distribution: Both sexes affected (autosomal). Affected individuals are infants/young children.

10. Diagnostics

Clinical / laboratory tests: - **Immunophenotyping (flow cytometry):** typically normal T and NK cell counts; **reduced B cells (89%)**; reduced memory B cells; arrest of B-cell maturation; reduced Treg/Tfr; elevated activation markers (PD-1, ICOS). - **Immunoglobulins:** impaired specific antibody responses; may trend toward hypogammaglobulinemia/agammaglobulinemia; **elevated IgE** in a subset (~22%). - **CBC: eosinophilia (~67%).** - **Functional assays:** reduced lymphocyte proliferation to anti-CD3; reduced NF- κ B activation ($\downarrow$ phospho-p65) on PMA/ionomycin or receptor stimulation; deficient IL-2/TNF- α production. These functional readouts are central to confirming pathogenicity.

Genetic testing (definitive): Molecular confirmation is required because the clinical picture overlaps with other CBM-opathies, DOCK8 deficiency, and hyper-IgE syndromes. - **Whole-exome / whole-genome sequencing (WES/WGS)** is the preferred first-line approach, especially given phenotypic overlap; upfront genomic sequencing shortens time to diagnosis in primary atopic disorders ([PMID: 39381601](#)). - **Targeted inborn-errors-of-immunity (IEI)/SCID/CID gene panels** including *MALT1*, *CARD11*, *BCL10*, *CARD9*, *DOCK8*. - **Single-gene *MALT1* sequencing** confirmed by Sanger; functional validation of variant impact strengthens classification.

Clinical criteria / differential diagnosis: No disease-specific consensus criteria exist; diagnosis rests on the combined immunodeficiency phenotype plus biallelic *MALT1* LOF with functional corroboration. **Differential diagnoses** to exclude: DOCK8 deficiency, hyper-IgE syndromes, other SCID/CID, BCL10 and CARD11 deficiencies (other CBM-opathies), IPEX, Omenn syndrome, and Wiskott-Aldrich syndrome. *"Although the presence of eczema, recurrent sinopulmonary, and staphylococcal infections are suggestive of DOCK8 deficiency, they are also a finding in CARD11 and MALT1 deficiency."* ([PMID: 39017781](#)). BCL10 deficiency shares an overlapping immunophenotype ([PMID: 34868072](#): *"this patient displays a reduction in NK, $\gamma\delta$ T, Tregs, and T"*).

Screening: No population newborn screening specifically detects MALT1 deficiency; notably, SCID TREC-based newborn screening may be normal because T-cell numbers are often preserved — an important caveat. **Cascade/carrier testing** in affected consanguineous families is valuable.

11. Outcome / Prognosis

Survival / mortality: Poor without curative therapy. In the largest cohort, *"Immunoglobulin replacement and antibiotics prophylaxis were mostly ineffective in reducing the frequency of infections and other complications. One patient received hematopoietic stem cell transplantation (HSCT) and five patients died as a complication of life-threatening infections."* (PMID: 35079916). The p.D471N patient experienced early death (PMID: 40748513).

Morbidity: High — recurrent severe infections, chronic enteropathy, failure to thrive, dermatitis, and autoimmune complications.

Disease-specific complications: Life-threatening bacterial/viral/fungal infections (leading cause of death), enteropathy with malnutrition, autoimmune manifestations.

Recovery potential: Minimal with supportive care alone; curative potential exists with successful allogeneic HSCT (reconstitutes CBM-competent hematopoietic cells).

Prognostic factors: Timing of diagnosis and access to HSCT; infection burden and organ damage at presentation; likely residual MALT1 function (hypomorphic vs. null variant).

12. Treatment

Supportive / pharmacotherapy (largely palliative): - **Immunoglobulin replacement (IVIG/ SCIG)** — NCIT: Intravenous Immunoglobulin Therapy. Largely ineffective at preventing infections in this disease. - **Antimicrobial / antifungal / antiviral prophylaxis.** Mostly ineffective at reducing complications. - **Management of enteropathy and dermatitis** — nutritional support, topical/skin care.

Supporting evidence: *"Immunoglobulin replacement and antibiotics prophylaxis were mostly ineffective in reducing the frequency of infections and other complications."* (PMID: 35079916).

Curative / advanced therapeutics: - **Allogeneic hematopoietic stem cell transplantation (HSCT)** — NCIT: Allogeneic Hematopoietic Stem Cell Transplantation (C15431). The **only established curative therapy**; reconstitutes CBM-competent lymphocytes. Reported in only a minority of patients to date; early transplantation before severe infectious damage is the goal. - **Gene therapy / gene editing:** Not yet available; conceptually attractive given the monogenic, hematopoietic-restricted nature of the defect — a plausible future direction but no clinical programs reported.

Pharmacologic caution — MALT1 inhibitors: MALT1 protease *inhibitors* are in development for lymphoma and autoimmune disease. Importantly, pharmacological MALT1 inhibition reproduces an **IPEX-like, Treg-depleting autoimmune pathology** in animal models ([PMID: 32425939](#)): *"pharmacological inhibition of MALT1 was associated with a rapid and dose-dependent reduction in Tregs and resulted in the progressive appearance of immune abnormalities and clinical signs of an IPEX-like pathology."* This underscores that *reducing* MALT1 function is deleterious — relevant both to understanding the patient phenotype and to avoiding iatrogenic harm.

Treatment outcomes / strategy: No formal treatment algorithms exist for this ultra-rare disease; management mirrors severe combined/combined immunodeficiency protocols (prophylaxis + definitive HSCT). Genotype-guided prognostication (null vs. hypomorphic) may refine urgency.

13. Prevention

- **Primary prevention:** Not possible for the monogenic disease itself. **Genetic counseling** for consanguineous families and carrier couples is the principal preventive measure; **preimplantation genetic diagnosis (PGD)** and **prenatal testing** are options when the familial variant is known.
- **Secondary prevention:** Early molecular diagnosis enables timely HSCT before irreversible infectious/organ damage. **Cascade screening** of at-risk relatives in affected families.
- **Tertiary prevention:** Infection prophylaxis, nutritional support, and vigilant management of autoimmune/atopic complications pending definitive therapy.
- **Immunization caveat:** As in other combined immunodeficiencies, **live vaccines are contraindicated**; vaccine responses are typically poor.

- **Public health:** Awareness of consanguinity-associated recessive IEI and incorporation of *MALT1* into IEI gene panels improve detection.
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14. Other Species / Natural Disease

Taxonomy / orthologs: *MALT1* is evolutionarily conserved in mammals. Mouse *Malt1* (NCBI Gene) is the principal ortholog used in disease modeling.

Natural disease in animals: No naturally occurring MALT1-deficiency disease is established in companion animals or wildlife (no prominent OMIA entry). However, **pharmacological MALT1 protease inhibition in rats and dogs** produces a progressive **IPEX-like pathology** with severe Treg reduction ([PMID: 32425939](#)), demonstrating cross-species conservation of the Treg-dependent dysregulation mechanism.

Comparative biology: The scaffold/protease duality and CBM–NF- κ B axis are conserved across mammals, making mouse models highly informative. **Zoonotic potential:** Not applicable (non-infectious genetic disease).

15. Model Organisms

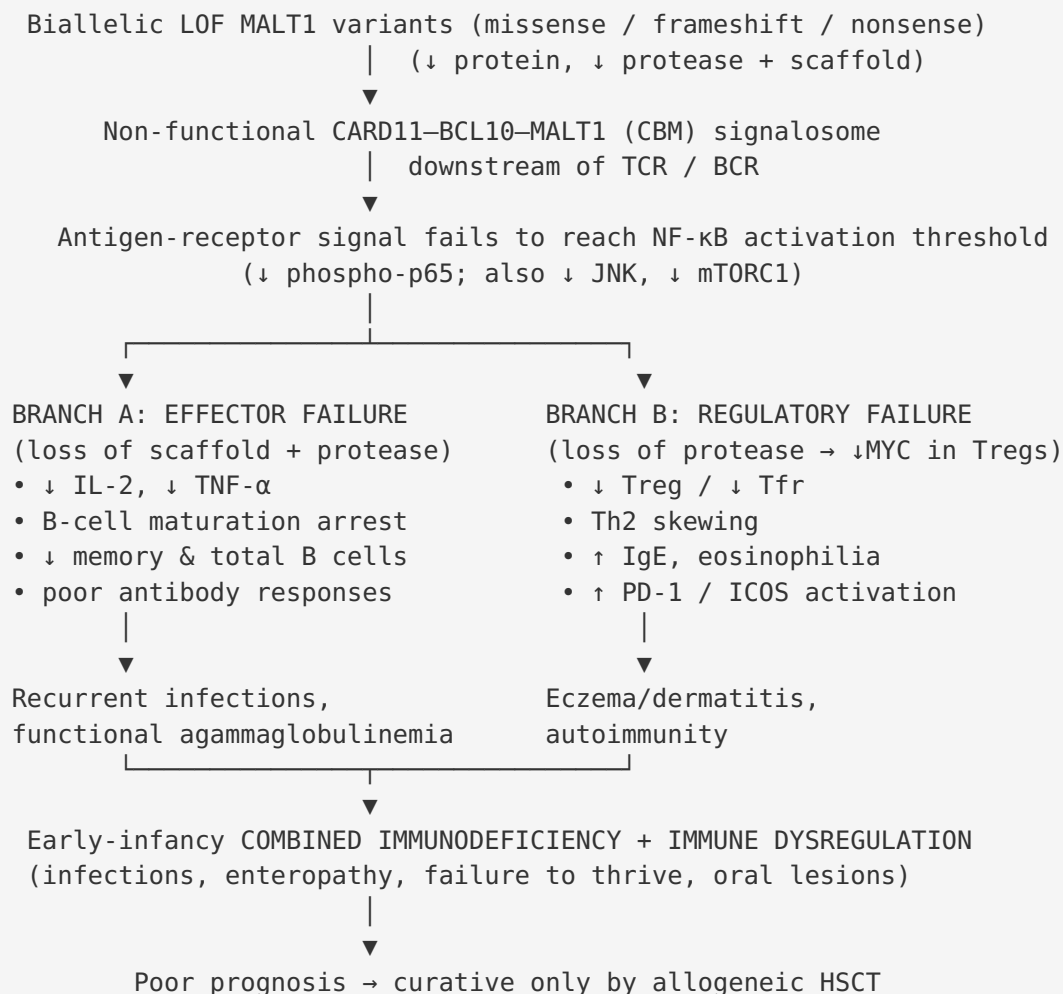
Mouse models are the principal system and recapitulate the immunodeficiency–dysregulation duality:

Model	Key phenotype	Reference
<i>Malt1</i> knockout (complete)	Atopic-like dermatitis upon aging; Th2 skewing; ↑ serum IgE; ↓ Treg frequency and CTLA-4	PMID: 31632405
Protease-dead (catalytically inactive, scaffold-retaining) MALT1 knock-in	Spontaneous autoimmunity from Treg loss and increased effector T-cell activation	PMID: 34668583
Treg-specific <i>Myc</i> deletion	Phenocopies lethal autoimmune syndrome (links MALT1 protease → MYC → Treg expansion)	PMID: 34668583
Pharmacological MALT1 inhibition (rat/dog)	Progressive IPEX-like pathology, severe Treg reduction	PMID: 32425939

Supporting quotes: - *"here we report that MALT1-deficient mice develop atopic-like dermatitis upon aging, which is preceded by Th2 skewing, an increase in serum IgE, and a decrease in Treg frequency and surface expression of the Treg functionality marker CTLA-4."* (PMID: 31632405) - *"MALT1 protease activity controls the TCR-induced upregulation of the transcription factor MYC and the subsequent expression of MYC target genes in Tregs"* (PMID: 34668583)

Phenotype recapitulation: Strong for the atopy/Th2/IgE/Treg-loss axis (dysregulation branch). **Limitations:** Complete-knockout mice emphasize dysregulation/atopy; the severe early-infancy infectious mortality seen in humans is less fully modeled, and the separation of scaffold vs. protease contributions (via protease-dead knock-ins) does not perfectly mirror human null/hypomorphic variants. **Model databases:** MGI, IMPC, IMSR.

Mechanistic Model (Synthesis)



The key unifying insight is that **MALT1's two biochemical activities map onto the two clinical faces of the disease**: scaffold-driven NF-κB loss drives effector/immunodeficiency features, while protease loss (via the MYC-mitochondrial axis in Tregs) drives the regulatory/dysregulation features. This is corroborated in patients (↓ NF-κB, ↓ IL-2/TNF-α, ↓ Treg/Tfr) and in model organisms (protease-dead knock-ins → Treg-dependent autoimmunity; pharmacologic inhibition → IPEX-like pathology).

Evidence Base

PMID	Title (abbrev.)	Contribution
35079916	<i>Expanding the Clinical and Immunological Phenotypes and Natural History of MALT1 Deficiency</i>	Largest cohort (n=19); phenotype frequencies, onset age, immunophenotype, poor outcome/HSCT
40748513	<i>Loss of MALT1 Function in a Patient With CID</i>	p.D471N LOF variant; impaired NF- κ B (↓ p-p65), ↓ IL-2/TNF- α , ↓ Treg/Tfr
42745540	<i>What Human MALT1 Deficiency Reveals About Immune Control</i>	CBM/NF- κ B threshold signaling framework
39017781	<i>A novel MALT1 variant (Egyptian patient)</i>	Frameshift p.Ile255TyrfsTer10; IMD12 designation; B-cell maturation arrest; DOCK8 differential
41882201	<i>Novel heterozygous variants in CARD11 and MALT1</i>	Hypomorphic p.K543R/p.M732T reduce enzymatic activity and stability
22158899	<i>Crystal structure of MALT1 paracaspase region</i>	Caspase-like fold, homodimerization, protease activity
21966355	<i>Oligomeric structure of MALT1 tandem Ig-like domains</i>	Ig-domain oligomerization in signaling
37126937	<i>Function and targeting of MALT1 paracaspase</i>	Dual scaffold/protease function; substrate cleavage (A20, CYLD, RelB, Regnase-1, Roquin)
31632405	<i>MALT1-Deficient Mice Develop Atopic-Like Dermatitis</i>	KO mouse: atopy, Th2, ↑ IgE, ↓ Treg/CTLA-4
34668583	<i>MALT1 protease → MYC in Tregs</i>	Protease–MYC–mitochondrial axis for Treg expansion
32425939	<i>Pharmacological MALT1 Inhibition → IPEX-Like Pathology</i>	Protease-specific loss → Treg-dependent autoimmunity (rat/dog)
30283440	<i>The CBM-opathies</i>	Classification/spectrum of CARD11-BCL10-MALT1 inborn errors
34868072	<i>Human BCL10 Deficiency</i>	Overlapping immunophenotype (↓ NK, $\gamma\delta$ T, Treg, T) in the differential

Evidence source types: Human clinical (cohorts/case reports: 35079916, 40748513, 39017781, 41882201, 34868072); in vitro/structural (22158899, 21966355, 37126937); model organism (31632405, 34668583, 32425939).

Limitations and Knowledge Gaps

1. **Extreme rarity ($n \approx 19$ + scattered case reports).** All epidemiology is case-derived; no validated prevalence/incidence, penetrance, or expressivity estimates. Phenotype frequencies may be biased by ascertainment toward severe/consanguineous cases.
 2. **Genotype–phenotype correlation is incompletely resolved.** The spectrum of variant types (null vs. hypomorphic) and how residual activity modulates the immunodeficiency-vs-dysregulation balance remain under-characterized.
 3. **HSCT outcome data are sparse** — only a minority of reported patients were transplanted; optimal conditioning, timing, and long-term outcomes are not established.
 4. **Human mechanistic data are limited to a few patients;** much of the protease-vs-scaffold dissection relies on model organisms and pharmacology, which may not perfectly mirror human null/hypomorphic variants.
 5. **No disease-specific QoL, biomarker-based prognostic, or newborn-screening data.** Standard SCID TREC screening may miss MALT1 deficiency because T-cell numbers are often preserved.
 6. **No gene therapy** programs or formal treatment algorithms exist.
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Proposed Follow-up Experiments / Actions

1. **Establish an international MALT1-deficiency patient registry** to aggregate genotype, immunophenotype, treatment, and HSCT outcome data and derive robust natural-history and genotype–phenotype correlations.
2. **Systematic functional classification of *MALT1* variants** (NF- κ B reporter, protease-cleavage, protein-stability assays) to build an ACMG-aligned variant interpretation resource distinguishing null vs. hypomorphic alleles.

3. **Define HSCT best practices** (timing, conditioning, donor source) through multicenter outcome analysis, and evaluate whether early transplantation (before severe infection) improves survival.
 4. **Single-cell transcriptomic/immune profiling of patient PBMCs** to map, at cell-type resolution, the effector vs. regulatory lesions (B-cell maturation block, Treg/Tfr deficits) and identify candidate biomarkers.
 5. **Preclinical evaluation of gene correction** (lentiviral *MALT1* addition or base/prime editing in patient HSPCs) given the monogenic, hematopoietic-restricted nature of the defect.
 6. **Incorporate *MALT1* into standard IEI/SCID gene panels** and raise clinician awareness of the DOCK8/hyper-IgE phenotypic overlap and the SCID-screening blind spot (preserved T cells).
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Report compiled from autonomous multi-iteration literature synthesis (8 confirmed findings, 25 papers reviewed). All clinical and mechanistic claims are attributed to the cited primary literature by PMID.
